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\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\localizer
TA: 19 sec Coil Selection: Auto Voxel Size: 0.5×0.5×10.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	On
Load Images to Graphic Segments	On
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
TR	20.0 ms
TE	5.00 ms
Averages	1
Concatenations	7
AutoAlign	---

Contrast - Common

TR	20.0 ms
TE	5.00 ms
TD	0.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	40 deg

Contrast - Common

Fat-Water Contrast	Standard
Dark Blood	Off
Contrasts	1
SWI	Off
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Multiple Series	Each Measurement

Resolution - Common

FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
Base Resolution	256
Phase Resolution	50 %
Interpolation	On

Resolution - Acceleration

Acceleration Mode	None
Deep Resolve	Off
Phase Partial Fourier	Off
Asymmetric Echo	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	On
POCS	Off
Distortion Correction	2D
Normalize	Off
Image Filter	Off

Geometry - Common

Slice Group	1
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Slices	5
Distance Factor	20 %
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Slices	5
Distance Factor	20 %
Position	Isocenter

Geometry - Common

Orientation	Sagittal
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	280 mm
FOV Phase	100.0 %
Slice Thickness	10.0 mm
TR	20.0 ms
Multi-Slice Mode	Sequential
Series	Interleaved
Concatenations	7

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	2
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
Slice Group	3
Position	Isocenter
Orientation	Sagittal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Sagittal
Initial Rotation	0.00 deg

Geometry - Saturation

Saturation Mode	Standard
Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	L >> R
Coronal	P >> A
Transversal	F >> H
Coil Combination	Adaptive Combine
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Tune up
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	263 mm
R >> L	350 mm
F >> H	350 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Slice-sel.
LR Balancing	Off

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	20.0 ms
Segments	1
Concatenations	7

Physio - Cardiac

Tagging	None
Fat-Water Contrast	Standard
Magn. Preparation	None
Dark Blood	Off
FOV Read	280 mm
FOV Phase	100.0 %
Phase Resolution	50 %

Physio - PACE

Resp. Control	Off
Concatenations	7

Inline - Liver

Liver Registration	Off
Save Original Images	On

Inline - Subtraction

Subtract	Off
Measurements	1

Inline - Subtraction

StdDev	Off
Save Original Images	On

Inline - MIP

MIP Sag	Off
MIP Cor	Off
MIP Tra	Off
MIP Time	Off
Radial MIP	Off
Save Original Images	On
MPR Sag	Off
MPR Cor	Off
MPR Tra	Off

Inline - Soft Tissue

Wash-in	Off
Wash-out	Off
TTP	Off
PEI	Off
MIP Time	Off
Measurements	1

Inline - Composing

Inline Composing	Off

Inline - MapIt

MapIt	None
Flip Angle	40 deg
Measurements	1
Contrasts	1
TE	5.00 ms
TR	20.0 ms
Save Original Images	On

Sequence - Part 1

Sequence Name	fl
Dimension	2D
Excitation	Slice-sel.
RF Pulse Type	Fast
Gradient Mode	Normal
Flow Compensation	None
Bandwidth	180 Hz/Px
Asymmetric Echo	Off
Segments	1

Sequence - Part 2

Introduction	On
RF Spoiling	On
Acoustic noise reduction	Off

Sequence - Assistant

SAR Assistant	Off
Allowed Delay	0 s

\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\Slice positioning
TA: 23 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	98.2 %
Slice Thickness	2.0 mm
TR	750.0 ms
TE	25.00 ms
Averages	1
Multi-band accel. factor	6
AutoAlign	Head > Brain

Contrast - Common

TR	750.0 ms
TE	25.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	52 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	1
Delay in TR	0.00 ms

Resolution - Common

FOV Read	220 mm
FOV Phase	98.2 %
Slice Thickness	2.0 mm
Base Resolution	110
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference scan mode	Single-shot
Acceleration Factor PE	2
Reference Lines PE	32
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Prescan

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	98.2 %
Slice Thickness	2.0 mm
TR	750.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	6

Geometry - AutoAlign

Slice Group	1
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	R0.3 A14.1 F41.0
R	0.3 mm
A	14.1 mm
F	41.0 mm
Initial Orientation	T > C
T > C	-16.10
> S	-0.50
Initial Rotation	179.01 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Rotation	179.01 deg
A >> P	216 mm
R >> L	220 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	750.0 ms
Log Signals	Off
Multi-band accel. factor	6

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	1
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance*
Flow Compensation	None
Bandwidth	2272 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	108

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	3200 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On

Sequence - Special

MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\BOLD 750ms mb6 gr2

TA: 2:52 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	98.2 %
Slice Thickness	2.0 mm
TR	750.0 ms
TE	25.00 ms
Averages	1
Multi-band accel. factor	6
AutoAlign	Head > Brain

Contrast - Common

TR	750.0 ms
TE	25.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	52 deg
Fat-Water Contrast	Fat Saturation
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	200
Delay in TR	0.00 ms

Resolution - Common

FOV Read	220 mm
FOV Phase	98.2 %

Resolution - Common

Slice Thickness	2.0 mm
Base Resolution	110
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference scan mode	Single-shot
Acceleration Factor PE	2
Reference Lines PE	32
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Prescan

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	98.2 %
Slice Thickness	2.0 mm
TR	750.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	6

Geometry - AutoAlign

Slice Group	1
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	R0.3 A14.1 F41.0
R	0.3 mm
A	14.1 mm
F	41.0 mm
Initial Orientation	T > C
T > C	-16.10
> S	-0.50
Initial Rotation	179.01 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Rotation	179.01 deg
A >> P	216 mm
R >> L	220 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	750.0 ms
Log Signals	Off
Multi-band accel. factor	6

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	200
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance*
Flow Compensation	None
Bandwidth	2272 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	108

Sequence - Part 2

Introduction	Off
RF Spoiling	Off

Sequence - Special

Excite pulse duration	3200 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On

Sequence - Special

MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\BOLD 750ms mb6 gr2 5echo
TA: 1:55 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: 2 Rel. SNR: 1.00

Properties

Start measurement without further preparation	Off
Wait for User to Start	On
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	98.2 %
Slice Thickness	2.0 mm
TR	2500.0 ms
TE 1	25.00 ms
TE 2	57.06 ms
TE 3	89.10 ms
TE 4	121.16 ms
TE 5	153.20 ms
Averages	1
Multi-band accel. factor	6
AutoAlign	Head > Brain

Contrast - Common

TR	2500.0 ms
TE 1	25.00 ms
TE 2	57.06 ms
TE 3	89.10 ms
TE 4	121.16 ms
TE 5	153.20 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Fat-Water Contrast	Fat Saturation
Contrasts	5
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	20
Delay in TR	0.00 ms

Resolution - Common

FOV Read	220 mm
FOV Phase	98.2 %
Slice Thickness	2.0 mm
Base Resolution	110
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	GRAPPA
Reference scan mode	Single-shot
Acceleration Factor PE	2
Reference Lines PE	32
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Prescan

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	P >> A
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	98.2 %
Slice Thickness	2.0 mm
TR	2500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	6

Geometry - AutoAlign

Slice Group	1
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	P >> A
AutoAlign	Head > Brain
Initial Position	R0.3 A14.1 F41.0

Geometry - AutoAlign

R	0.3 mm
A	14.1 mm
F	41.0 mm
Initial Orientation	T > C
T > C	-16.10
> S	-0.50
Initial Rotation	179.01 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	0 mm
Table Position	H
Inline Composing	Off

System - Miscellaneous

Coil Selection	ACS All but spine
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Rotation	179.01 deg
A >> P	216 mm
R >> L	220 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off

System - Tx/Rx

Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	2500.0 ms
Log Signals	Off
Multi-band accel. factor	6

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	20
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Baseline
Meas[4]	Baseline
Meas[5]	Baseline
Meas[6]	Baseline
Meas[7]	Baseline
Meas[8]	Baseline
Meas[9]	Baseline
Meas[10]	Baseline
Meas[11]	Active
Meas[12]	Active
Meas[13]	Active
Meas[14]	Active
Meas[15]	Active
Meas[16]	Active
Meas[17]	Active
Meas[18]	Active
Meas[19]	Active
Meas[20]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	20
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epfid
Dimension	2D
Excitation	Standard
Gradient Mode	Performance*
Flow Compensation	None
Bandwidth	2272 Hz/Px
Echo Spacing	0.56 ms
Free Echo Spacing	Off
EPI Factor	108

Sequence - Part 2

Introduction	Off

Sequence - Part 2

RF Spoiling	Off

Sequence - Special

Excite pulse duration	3200 us
Min. prep scans	0
Min. prep scans SB	0
Inter-TE delay	0 us
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Echoes in separate series	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\cmrr_mbep2d_se

TA: 36 sec Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
TE	71.60 ms
Averages	1
Multi-band accel. factor	6
AutoAlign	Head > Brain

Contrast - Common

TR	1500.0 ms
TE	71.60 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Enabled
Contrasts	1
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	15
Delay in TR	0.00 ms

Resolution - Common

FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	110
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	1500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	6

Geometry - AutoAlign

Slice Group	1
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.3 A14.1 F41.0
R	0.3 mm
A	14.1 mm
F	41.0 mm
Initial Orientation	T > C
T > C	-16.10
> S	-0.50
Initial Rotation	-0.99 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	41 mm
Table Position	F
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Rotation	-0.99 deg
A >> P	220 mm
R >> L	220 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	1500.0 ms
Log Signals	Off
Multi-band accel. factor	6

BOLD

GLM Statistics	Off
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	Off
Spatial Filter	Off
Measurements	15
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	2392 Hz/Px
Echo Spacing	0.54 ms
Free Echo Spacing	Off
EPI Factor	110

Sequence - Part 2

Introduction	Off

Sequence - Special

Excite pulse duration	3200 us
Refocus pulse duration	6400 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
Time-shifted MB RF	On
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Force GPA balance	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\cmrr_mbep2d_se_3echo
TA: 1:17 min Coil Selection: Auto Voxel Size: 2.0×2.0×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	3500.0 ms
TE 1	71.60 ms
TE 2	143.32 ms
TE 3	215.04 ms
Averages	1
Multi-band accel. factor	6
AutoAlign	Head > Brain

Contrast - Common

TR	3500.0 ms
TE 1	71.60 ms
TE 2	143.32 ms
TE 3	215.04 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Enabled
Contrasts	3
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Measurements	15

Contrast - Dynamic

Delay in TR	0.00 ms

Resolution - Common

FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
Base Resolution	110
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	Off

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Hamming	Off
Distortion Correction	2D
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	220 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	3500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	6

Geometry - AutoAlign

Slice Group	1
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.3 A14.1 F41.0
R	0.3 mm
A	14.1 mm
F	41.0 mm
Initial Orientation	T > C
T > C	-16.10

Geometry - AutoAlign

> S	-0.50
Initial Rotation	-0.99 deg

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	41 mm
Table Position	F
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Rotation	-0.99 deg
A >> P	220 mm
R >> L	220 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3500.0 ms

Physio - Signal

Log Signals	Off
Multi-band accel. factor	6

BOLD

GLM Statistics	On
Ignore Meas. at Start	0
Ignore After Transition	0
Model Transition States	On
Temp. Highpass Filter	On
Threshold	4.00
Paradigm Size	3
Meas[1]	Baseline
Meas[2]	Baseline
Meas[3]	Active
Motion Correction	On
Spatial Filter	Off
Measurements	15
Delay in TR	0.00 ms

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	2392 Hz/Px
Echo Spacing	0.54 ms
Free Echo Spacing	Off
EPI Factor	110

Sequence - Part 2

Introduction	Off

Sequence - Special

Excite pulse duration	3200 us
Refocus pulse duration	6400 us
Min. prep scans	0
Min. prep scans SB	0
Inter-TE delay	0 us
Delay before PC scans	0 us
Single-band images	On
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	On
Opt. MB RF pulse BW	Off
Time-shifted MB RF	On
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
Echoes in separate series	Off
Force GPA balance	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off
Online multi-band recon.	Online
FFT scale factor	1.00

Sequence - Special

Physio recording	Off
Triggering scheme	Standard

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\cmrr_mbep2d_diff
TA: 28 sec Coil Selection: Auto Voxel Size: 3.9×3.9×2.0 mm³ Acc:: None Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	3500.0 ms
TE	200.00 ms
Multi-band accel. factor	6
AutoAlign	Head > Brain

Contrast - Common

TR	3500.0 ms
TE	200.00 ms
MTC	Off
Magn. Preparation	None
Flip Angle	90 deg
Refocus flip angle	180 deg
Fat-Water Contrast	Fat Saturation
Grad. rev. fat suppr.	Enabled
Reconstruction	Magnitude

Contrast - Dynamic

Dynamic Mode	Standard
Multiple Series	Off
Delay in TR	0.00 ms

Resolution - Common

FOV Read	500 mm
FOV Phase	100.0 %

Resolution - Common

Slice Thickness	2.0 mm
Base Resolution	128
Phase Resolution	100 %
Interpolation	Off

Resolution - Acceleration

Acceleration Mode	None
Phase Partial Fourier	6/8

Resolution - Filter

Raw Filter	Off
Elliptical Filter	Off
Distortion Correction	Off
Static Field Correction	Off
Normalize	Off

Geometry - Common

Slice Group	1
Slices	72
Distance Factor	0 %
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	A >> P
Phase Oversampling	0 %
FOV Read	500 mm
FOV Phase	100.0 %
Slice Thickness	2.0 mm
TR	3500.0 ms
Multi-Slice Mode	Interleaved
Series	Interleaved
Multi-band accel. factor	6

Geometry - AutoAlign

Slice Group	1
Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Phase Encoding Dir.	A >> P
AutoAlign	Head > Brain
Initial Position	R0.3 A14.1 F41.0
R	0.3 mm
A	14.1 mm
F	41.0 mm
Initial Orientation	T > C
T > C	-16.10
> S	-0.50
Initial Rotation	-0.99 deg

Geometry - Navigator

Geometry - Saturation

Special Saturation	None

Geometry - Tim Planning Suite

Set-n-Go Protocol	Off
Table Position	41 mm
Table Position	F
Inline Composing	Off

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H
Coil Combination	Sum of Squares
Matrix Optimization	Off
Coil Focus	Flat

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Standard
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off

System - Adjust Volume

Position	R0.3 A14.1 F41.0 mm
Orientation	T > C-16.1 > S-0.5
Rotation	-0.99 deg
A >> P	500 mm
R >> L	500 mm
F >> H	144 mm
Reset	Off

System - pTx

B1 Shim	TrueForm
Excitation	Standard

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3500.0 ms
Multi-band accel. factor	6

Physio - PACE

Resp. Control	Off

Physio - PACE

Multi-band accel. factor	6

Diff

Diffusion Mode	MDDW
Diff. Directions	6
Diffusion Scheme	Bipolar
Diff. Weightings	1
b-value	0 s/mm²
Averages	1
Dynamic Field Correction	Off
Invert Gray Scale	Off
Diff. Weighted Images	On
Trace Weighted Images	Off
Tensor	Off
FA Maps	Off
ADC Maps	Off
Exponential ADC Maps	Off
Calculated Image	Off

Sequence - Part 1

Sequence Name	epse
Dimension	2D
Excitation	Standard
Gradient Mode	Performance
Bandwidth	1502 Hz/Px
Echo Spacing	0.93 ms
Free Echo Spacing	Off
EPI Factor	128

Sequence - Part 2

Introduction	Off
RF Spoiling	Off
Phase Correction	Internal

Sequence - Special

Excite pulse duration	2560 us
Refocus pulse duration	5120 us
Min. prep scans	0
Min. prep scans SB	0
Delay before PC scans	0 us
Single-band images	Off
MB LeakBlock kernel	On
MB dual kernel	Off
MB RF phase scramble	Off
Opt. MB RF pulse BW	Off
Time-shifted MB RF	Off
SENSE1 coil combine	Off
Invert RO/PE polarity	Off
PF omits higher k-space	Off
Force GPA balance	Off
Disable B1 control loop	Off
Disable freq. update	Off
Suppress 16-bit DICOM	On
Force equal slice timing	Off

Sequence - Special

Online multi-band recon.	Online
FFT scale factor	1.00
Physio recording	Off

Sequence - Assistant

SAR Assistant	Off

\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\eja_svs_laser
TA: 3:39 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	laser

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Spoiling scheme	0
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Rect excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\eja_svs_mpress
TA: 6:51 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	mpres

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	1.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
MEGA flip angle	180 deg
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
PRESS+4	Off
Symm. MEGA timing	On
MEGA water suppr.	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	9.00 ppm

Sequence - Special

Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	100.00 Hz

\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\eja_svs_mslaser
TA: 6:51 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	mslsr

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
MEGA flip angle	180 deg
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	0 deg
OVS flip angle SL	0 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
MEGA water suppr.	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Asym. excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	9.00 ppm

Sequence - Special

Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	100.00 Hz

\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\eja_svs_press
TA: 3:39 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	press

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\eja_svs_steam
TA: 3:39 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
TM	75.00 ms
Flip Angle	90 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	steam

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

RF pulse duration	2560 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	1.50
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
Symmetric RF pulses	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\eja_svs_slaser
TA: 3:39 min Coil Selection: Auto Vol: 20×20×20 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	64

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	64
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
Vol A >> P	20 mm
Vol R >> L	20 mm
Vol F >> H	20 mm

Geometry - AutoAlign

AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm
F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	slasr

Sequence - Common

Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	0 deg
OVS flip angle SL	0 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Resolve averages	Off
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Asym. excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\eja_csi_laser
TA: 13:09 min Coil Selection: Auto Voxel Size: 12.5×12.5×20.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	1
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Hamming	Off
Dimension	2D
Phase Encoding	Full

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm

System - Adjust Volume

F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	laser
Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Dimension	2D
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Spoiling scheme	0
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
GOIA refoc. pulses	Off

Sequence - Special

Rect excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\eja_csi_mslaser
TA: 25:57 min Coil Selection: Auto Voxel Size: 12.5×12.5×20.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	1
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Hamming	Off
Dimension	2D
Phase Encoding	Full

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm

System - Adjust Volume

F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	mslsr
Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Dimension	2D
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
MEGA flip angle	180 deg
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	0 deg
OVS flip angle SL	0 deg
VAPOR delay 6	68 ms

Sequence - Special

VAPOR delay 5	106 ms
VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
MEGA water suppr.	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Asym. excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None
Number of label freqs.	2
Editing pulse freq. [1]	9.00 ppm
Editing pulse freq. [2]	1.90 ppm
Editing pulse BW	100.00 Hz

\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\eja_csi_press
TA: 13:09 min Coil Selection: Auto Voxel Size: 12.5×12.5×20.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	1
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Hamming	Off
Dimension	2D
Phase Encoding	Full

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm

System - Adjust Volume

F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	press
Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Dimension	2D
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	2600 us
Refocus pulse duration	5200 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
Forced min. TE1	0 ms
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	90 deg
OVS flip angle SL	90 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms
VAPOR delay 4	105 ms

Sequence - Special

VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
'RR' refoc. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None

\\Research\Investigators - validated on FIT\Frederick\CMRR test scans\eja_csi_slaser
TA: 13:09 min Coil Selection: Auto Voxel Size: 12.5×12.5×20.0 mm³ Rel. SNR: 1.00

Properties

Start measurement without further preparation	On
Wait for User to Start	Off
Start measurements	Single Measurement
Prio Recon	Off
Auto Open Inline Display	Off
Auto Close Inline Display	Off
Load Images to MR View&GO	On
Auto Store Images	On
Disable auto transfer to PACS	Off
Load Images to Stamp Segments	Off
Load Images to Graphic Segments	Off
Graphic segment	Default
Inline Movie	Off

Routine

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm
TR	3000.0 ms
TE	135.00 ms
Averages	1

Contrast - Common

TR	3000.0 ms
TE	135.00 ms
Excite flip angle	90 deg
Refocus flip angle	180 deg
Preparation Scans	4
Averages	1
VAPOR	Water Saturation
VAPOR suppr.	Water Suppr.
Water s. BW	135.00 Hz
Water s. Delta Pos.	0.00 ppm

Resolution - Common

FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Scan Res. A >> P	16
Scan Res. R >> L	16
Interpol. Res. A >> P	16
Interpol. Res. R >> L	16
Hamming	Off
Dimension	2D
Phase Encoding	Full

Resolution - Common

Vector Size	2048
Normalize	Off

Geometry - Common

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
FOV A >> P	200 mm
FOV R >> L	200 mm
Thickness F >> H	20 mm
Vol A >> P	20 mm
Vol R >> L	20 mm

Geometry - AutoAlign

Slice Group	1
Position	Isocenter
Orientation	Transversal
Phase Encoding Dir.	A >> P
AutoAlign	---
Initial Position	Isocenter
L	0.0 mm
P	0.0 mm
H	0.0 mm
Initial Orientation	Transversal
Initial Rotation	0.00 deg

System - Miscellaneous

Coil Selection	Auto Coil Select
Radial Sorting	Off
MSMA	S - C - T
Sagittal	R >> L
Coronal	A >> P
Transversal	F >> H

System - Adjustments

Adjustment Strategy	Standard
B0 Shim	Advanced
B1 Shim	TrueForm
Adjustment Tolerance	Auto
Adjust with Body Coil	Off
Confirm Frequency	Never
Assume Silicone	Off
Adj. Water Suppr.	Off

System - Adjust Volume

Position	Isocenter
Orientation	Transversal
Rotation	0.00 deg
A >> P	20 mm
R >> L	20 mm

System - Adjust Volume

F >> H	20 mm
Reset	Off

System - pTx

B1 Shim	TrueForm

System - Tx/Rx

Frequency 1H	123.253118 MHz
? Ref. Amplitude 1H	0.000 V
Reset	Off
Image Scaling	1.000

Physio - Signal

1st Signal/Mode	None
TR	3000.0 ms

Sequence - Common

Sequence Name	slasr
Preparation Scans	4
Delta Frequency	-1.70 ppm
Measurements	1
Dimension	2D
Phase Cycling	Auto
Save Uncombined	Off
Bandwidth	2000 Hz
Acquisition Duration	1024 ms
Remove Oversampling	On

Sequence - Special

Excite pulse duration	8960 us
Refocus pulse duration	8960 us
Spoiler max. amplitude	20.00 mT/m
Spoiler amp. ratio	3.00
Refocus grad. factor	1.00
OVS slab thickness	80.00 mm
OVS slab pos. offset	5.00 mm
OVS delta frequency	0.00 ppm
Spoiler duration	1300 us
Acq. window shift	200 us
Min. settling delay	300 us
Gradient ramp time	140 us
HS refoc. pulse N	1
HS refoc. pulse R	20
VAPOR flip angle	80 deg
VAPOR delay 8	28 ms
VAPOR delay 7	74 ms
OVS pulse duration	5120 us
OVS HS pulse N	1
OVS HS pulse R	40
OVS flip angle RO	90 deg
OVS flip angle PH	0 deg
OVS flip angle SL	0 deg
VAPOR delay 6	68 ms
VAPOR delay 5	106 ms

Sequence - Special

VAPOR delay 4	105 ms
VAPOR delay 3	146 ms
VAPOR delay 2	100 ms
VAPOR delay 1	150 ms
Enable OVS	On
Use expt. SAR calc	Off
Check RF clipping	Off
Send ref. scans	Off
Inversion pulse	Off
GOIA refoc. pulses	Off
Asym. excit. pulse	Off
Invert SS grad. pol.	Off
Shift RO frequency	Off
Debug loop type	None